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MIL-G-62743(AT)
17 September 1992

MILITARY SPECIFICATION

GENERATOR, ENGINE ACCESSORY, 300 AMP

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance, test, manufacture and acceptance requirements for the 300 ampere direct current engine accessory generator, part number 19207-12351001-2, herein referred to as the generator (see 6.6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards, form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2920
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SPECIFICATIONS

FEDERAL

P-D-220	- Detergent, General Purpose.

MILITARY

MIL-A-8625	- Anodic Coatings for Aluminum and Aluminum Alloys.
MIL-I-46058	- Insulating Compound, Electrical (for Coating Printed Circuit Assemblies).
MIL-P-116	- Preservation, Methods of.
MIL-S-19500 JANTX	- Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-275	- Printed Wiring for Electronic Equipment.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461A Notice 4	- Electromagnetic Interference Characteristics Requirements for Equipment, Subsystems and Systems.
MIL-STD-462 Notice 3	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1275	- Characteristics of 28 Volt DC Electrical Systems in Military Vehicles.
MIL-STD-2000	- Standard Requirements for Soldered Electrical and Electronic Assembly.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

19207-12328817	- Critical Item Product Function Specification for Generator, Engine Accessory, 300 Amp
19207-12351001	- Generator, 300 Amp DC
19207-10929031	- Gage, Involute Go Composite Plug
19207-10929032	- Gage, Involute Not Go Sector Plug

ARMY REGULATIONS

AR 611-201	- Enlisted Career MGT Fields and Military Occupational Specialties (MOS 63G10).
MRC-82304 MRC/SD-EP-82-027	- Nuclear Survivability Criteria for the IFV/CFV Systems (U) dated 9 September 1980 published by the U.S. Army Nuclear and Chemical Agency.

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article (see 6.3).

3.1.1 Preproduction inspection (see 4.4.1). When specified (see 6.2), generators shall be subjected to preproduction testing. Production tooling is not required in the manufacture of preproduction samples.

3.1.2 Initial production inspection (see 4.4.2). When specified (see 6.2), generators shall be subjected to initial production inspection to demonstrate the adequacy and suitability of the contractor's processes and procedures in achieving the performance inherent in the design. Generators shall be produced with equipment and procedures used in production.

3.2 Materials. Materials shall be in accordance with the drawings, parts lists, and other documents specified on Drawing 19207-12351001. Materials shall be uniform in quality and free from defects or imperfections.

3.2.1 Dissimilar metals. The generator shall be designed and fabricated to minimize galvanic corrosion in dissimilar metal couples as specified in MIL-STD-889.

3.3 Design and construction. The design and construction of the generator shall be as specified herein and in accordance with Drawing 19207-12351001.

3.3.1 Weight. Weight of the generator shall be not greater than 85 pounds.

3.3.2 Dimensions. Dimensions of the generator shall be in accordance with drawing 19207-12351001.

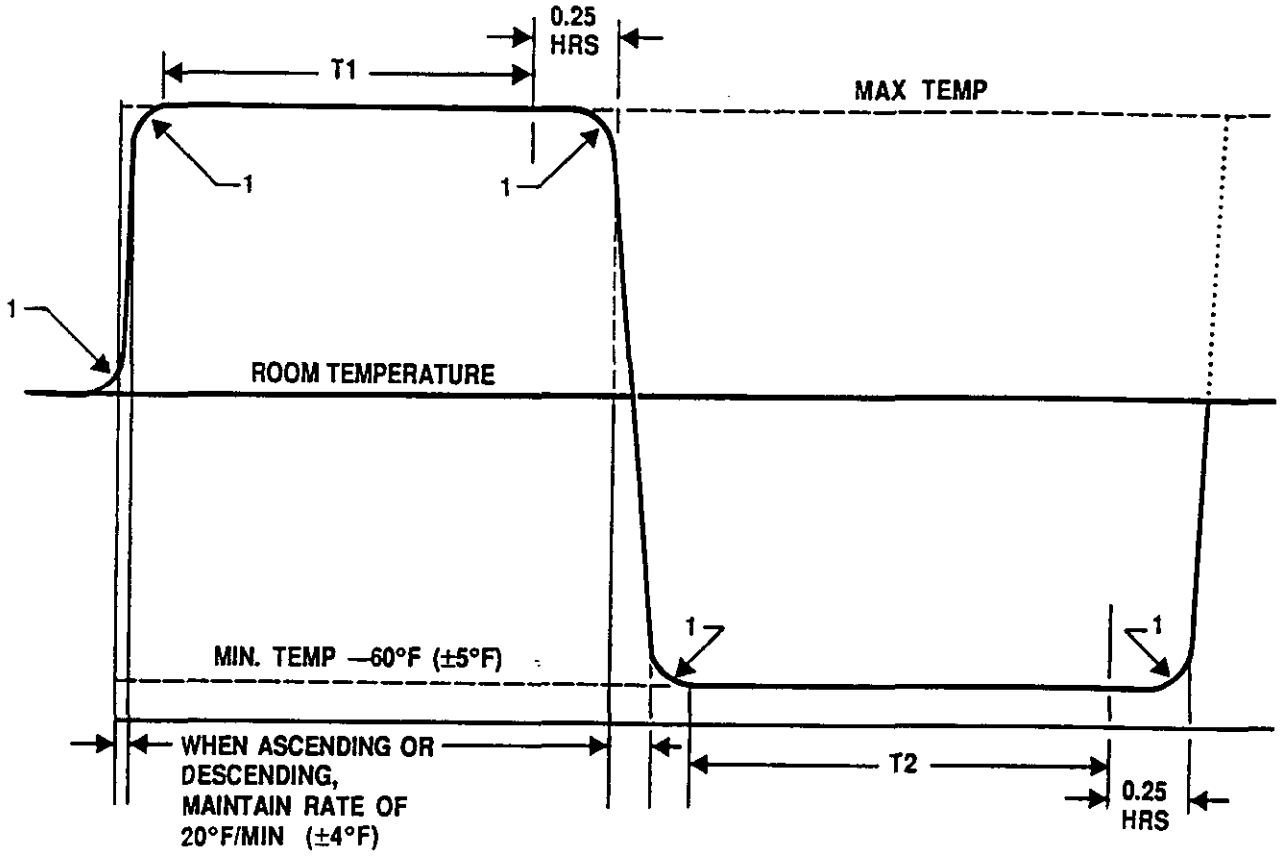
3.3.3 Maintainability requirements. Generator design shall accommodate repair capability at the lowest possible level of the Army Maintenance Operations using the Direct Support level as a goal. Consideration must be given to the available MOS 63G10 (AR 611-201) skills, tools, and test measurement and diagnostic equipment (TMDE). Requirement for, or introduction of, new TMDE is forbidden.

3.3.4 Spline alignment. Provision shall be made for means to rotate shaft for spline alignment.

3.3.5 Stress screen (see 4.6.2.10). The generator control unit(s) and the rectifier assembly shall each demonstrate no significant degradation in performance after being stress screened in accordance with figure 1. The generator control unit(s) shall be operated at nominal voltage for 15 cycles (see figure 1); the rectifier assembly shall be operated at a nominal voltage for 10 cycles (see figure 1).

3.3.6 Soldering. Soldering of electrical connections shall be in accordance with MIL-STD-2000, except solder used on power rectifier and output stud connections shall have a melting temperature not less than 200° Celsius (392 degrees Fahrenheit (°F)).

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NOTE: Control unit(s) upper temperature: not greater than 250°F +0 -5°F.

Rectifier upper temperature: not greater than 300°F +0 -5°F.

Rounding off at upper and lower limits allowable (not to exceed 20% of ramp temperature).

FIGURE 1. Stress screen cycle.

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3.3.7 Sealing of electronic assemblies (see 4.6.2.11). Potted assemblies shall have a vacuum applied during the potting process to ensure complete fill. The assembly shall be returned to atmospheric pressure while the potting compound is still fluid.

3.3.7.1 Conformal coating alternative. As an alternative to potting, the assembly may be conformally coated on both sides in accordance with the requirements of MIL-STD-275, using insulation compound type UR, MIL-I-46058, .001 to .010 inch thick, and sealed mechanically.

3.3.8 Bearings. The generator shall have sealed bearings. The generator shall require no lubrication when tested as specified in 3.5.11.

3.3.9 Guard. The generator shall be provided with a guard to protect personnel from the rotating parts.

3.3.10 Power rectifier (see 4.6.3.4). All diodes used in the power rectifier assembly shall be hermetically sealed and packaged in standard DO packages. It is the responsibility of the generator manufacturer to use conservative engineering judgement in selecting the proper device to insure reliable generator operation.

3.3.11 Power transistor. The power transistor controlling the field current of the generator shall conform to MIL-S-19500 JANTX.

3.3.12 Press fit assemblies (see 4.6.3.5). Components permanently joined by press fit shall be designed with a minimum holding force margin of 1.5. Holding forces shall be based on peak shock forces as specified in 3.6.3 and shall include the extreme high and low service temperature of the assembly.

3.3.13 Temperature-compensated regulator. The generator shall be equipped with a temperature-compensated regulator, the function of which is to provide the optimum voltage to recharge and maintain the vehicle batteries. The generator output voltage versus the generator mounted regulator temperature is shown in figure 2.

3.4 Protective coating.

3.4.1 Exterior finish. Exterior surfaces of the generator shall be painted as specified on Drawing 19207-12351001.

3.4.2 Aluminum components. Unless otherwise suitably protected, all aluminum components shall be anodized in accordance with MIL-A-8625.

3.5 Performance. Batteries connected, ambient temperature 65 to 85°F generator temperature stabilized unless specified. Unspecified tolerances are as follows: current, ± 10 amperes (amps); voltage, ± 1 volt direct current (vdc); speed, ± 500 revolutions per minute (rpm); temperature, ± 10°F.

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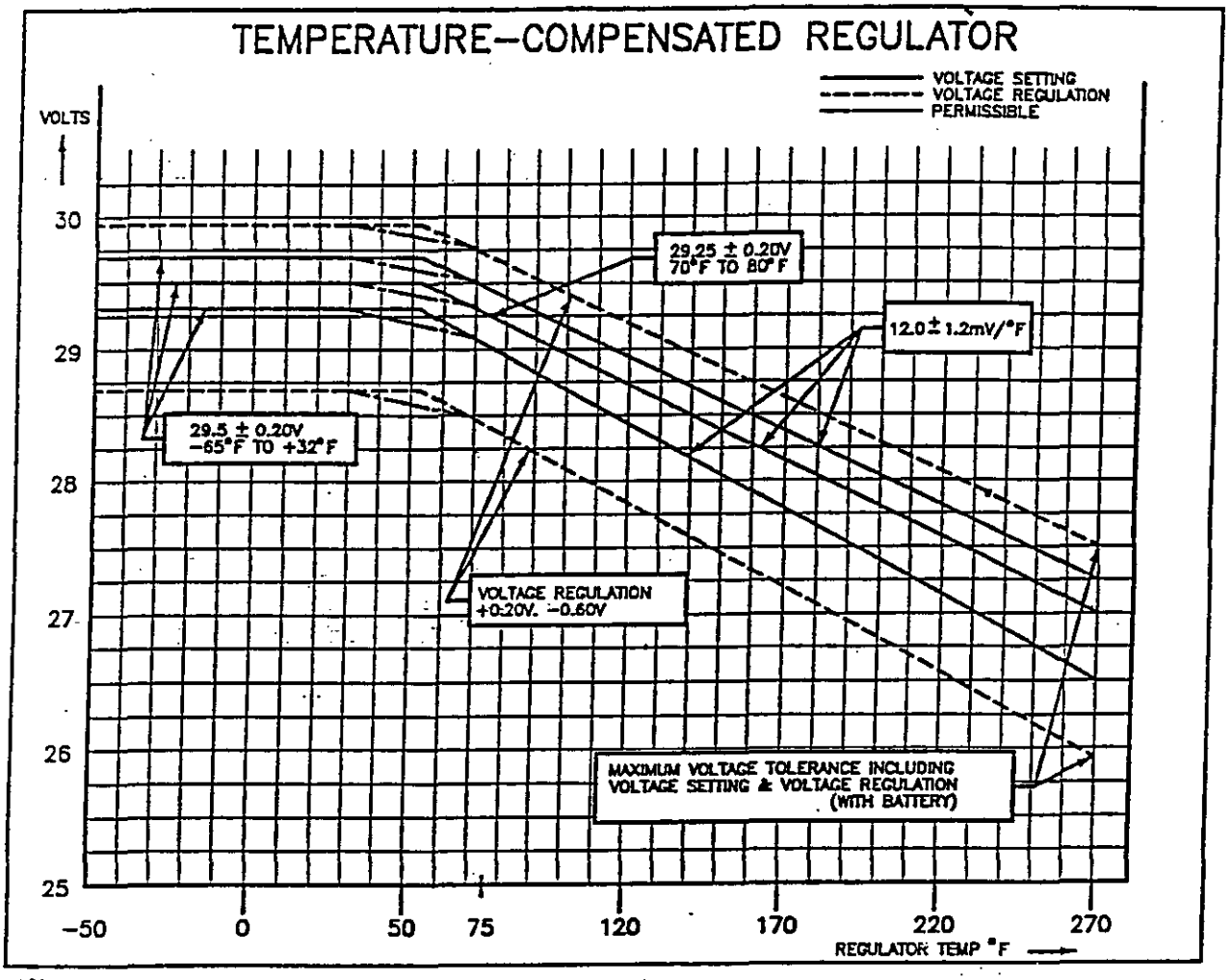


FIGURE 2. Temperature-compensated regulator.

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3.5.1 Generator output current (see 4.6.1.1). Minimum generator output current shall be as specified in table I. The generator output shall be enabled by the application of 18.0 to 30.0 vdc on the excitation contact and disabled by removal of this voltage. The current required to enable the generator shall be not greater than 1 amp. For each of the tests listed in table I, the minimum output voltage shall be 26.0 vdc.

TABLE I. Generator output current.

Test#	Minimum Rated continuous output current [amps] 1/	Maximum speed [rpm]	Generator air inlet Temp. [^{o}F]	Minimum test duration [hr]
1	120	2325	200° to 210°F	1
2	300	7500	75° to 100°F	1
3	260	7500	225° to 235°F	1

1/ Rated continuous output current shall be produced at maximum speed. It may be produced at any lower speed.

NOTE: Maximum current is defined herein as current equivalent to or greater than the rated continuous output current under the condition specified in table I.

3.5.2 Generator output voltage settings (see 4.6.1.2). The temperature-compensated voltage regulator shall exhibit voltage settings versus regulator temperature as specified in figure 2 when operated at 7500 ± 250 rpm, 50 ± 5 amp load. The generator shall be connected to a battery.

3.5.3 Voltage regulation (see 4.6.1.3). The generator shall use internal voltage sensing. Generator output voltage shall be stable or decrease with load increase. With battery, the voltage regulation tolerance shall be +0.20,-0.60 vdc over the speed, load, and temperature range specified in figure 1. Without battery, the voltage regulation tolerance shall be ± 1.0 vdc over the speed, load, and temperature ranges specified in figure 2.

3.5.4 Voltage ripple (see 4.6.1.4). The generator peak-to-peak (p-p) ripple voltage shall be not greater than 3.0 vdc over the operating range with battery connected and measurements taken at the battery terminals. Without battery in the steady-state condition, voltage peaks (including ripple) shall be not greater than 40.0 vdc over the operating range.

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3.5.5 Voltage transients (see 4.6.1.5). With battery on the line, the voltage transients shall be within the limits shown in figure 3. Without battery on the line, the voltage transients shall be within the limits shown in figure 4 (see 6.5). Load switching to produce overshoot is defined as 85 to 10 percent, and undershoot is defined as 10 to 85 percent of rated continuous current rating. The switching time is 20 milliseconds (ms) maximum.

3.5.6 Overvoltage protection (see 4.6.1.6). The generator shall have automatic shutdown to prevent a steady-state overvoltage condition in case of generator control failure. Voltage excursion normal to the system shall not cause generator shutdown. The overvoltage protection shall be built into the removable generator regulator subassembly.

3.5.6.1 Overvoltage shutdown with battery. The failed generator shall be turned off before the steady-state generator voltage reaches 38 vdc. In parallel generator operation, only the faulty generator shall shut down.

3.5.6.2 Overvoltage shutdown without battery. The failed generator shall be turned off before the voltage reaches 130 vdc peak. The generator voltage shall not exceed 100 vdc for more than 100 ms.

3.5.7 Isolated output terminal (see 4.6.1.7). The generator shall be equipped with an output terminal that is isolated from the generator B+ terminal to operate generator indicator lights and vehicle accessories. The output voltage shall be equal to the generator main output voltage while the generator is operational, and be reduced when the generator is turned off or malfunctions. This output is connected to terminal C on the energizing connector. The maximum current load is 1.5 amp at 230°F generator air inlet temperature.

3.5.8 Efficiency (see 4.6.1.8). Minimum generator efficiency above 100 amps in the 3000 to 7000 rpm range shall be 60 percent. Torque required to drive the generator at any speed up to 8500 rpm shall not exceed 30 foot-pounds.

3.5.9 Overspeed (see 4.6.1.9). Generator speeds up to 10,000 rpm for a period of up to 10 minutes shall not result in any damage or impairment of generator performance.

3.5.10 Maximum capacity (see 4.6.1.10). The generator shall be capable of protecting itself from damage when operated for periods of up to 6 hours loaded beyond the regulated output voltage at 230 ± 5°F at any speed between 2300 and 7500 rpm.

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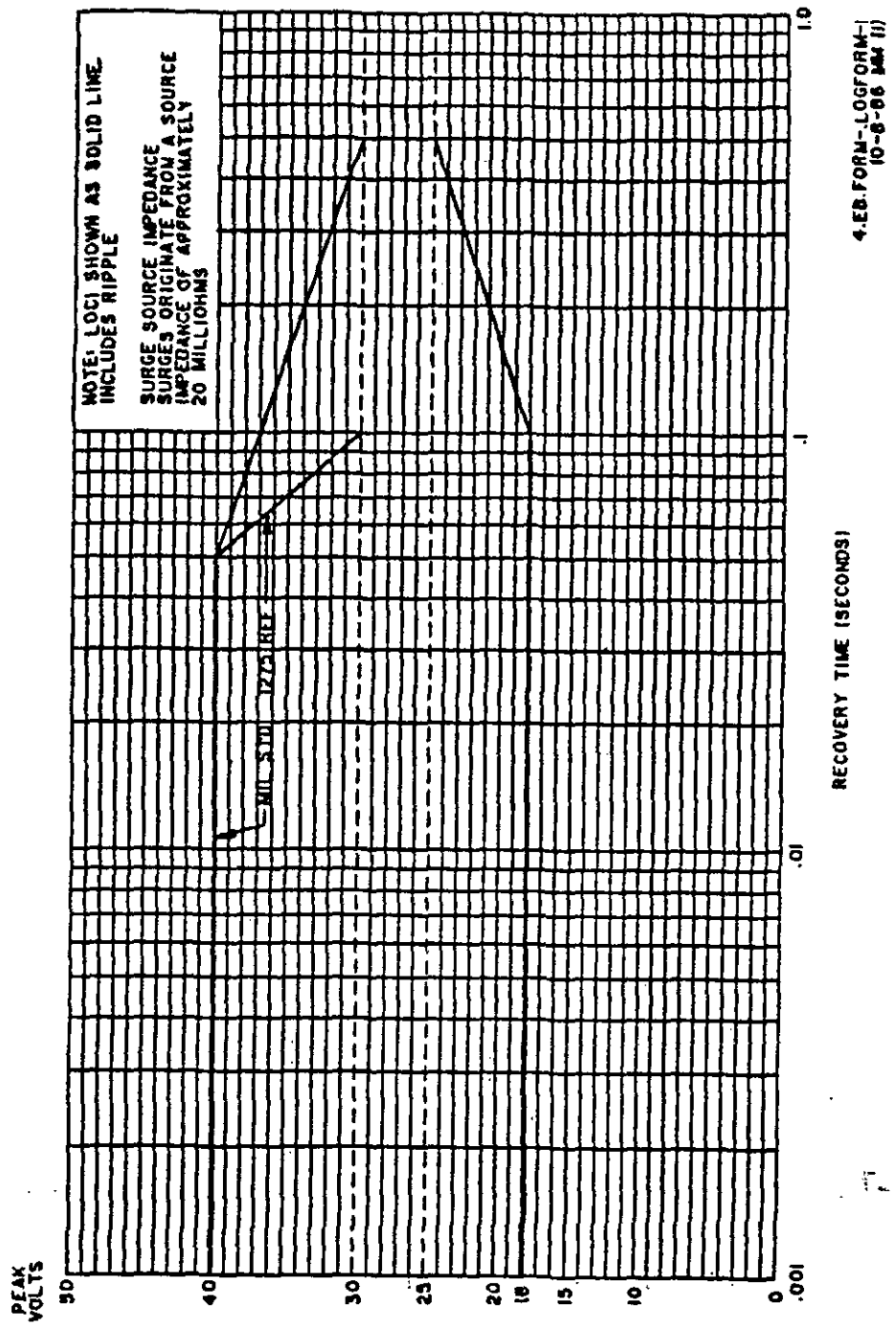


FIGURE 3. Fault-free condition, with battery.

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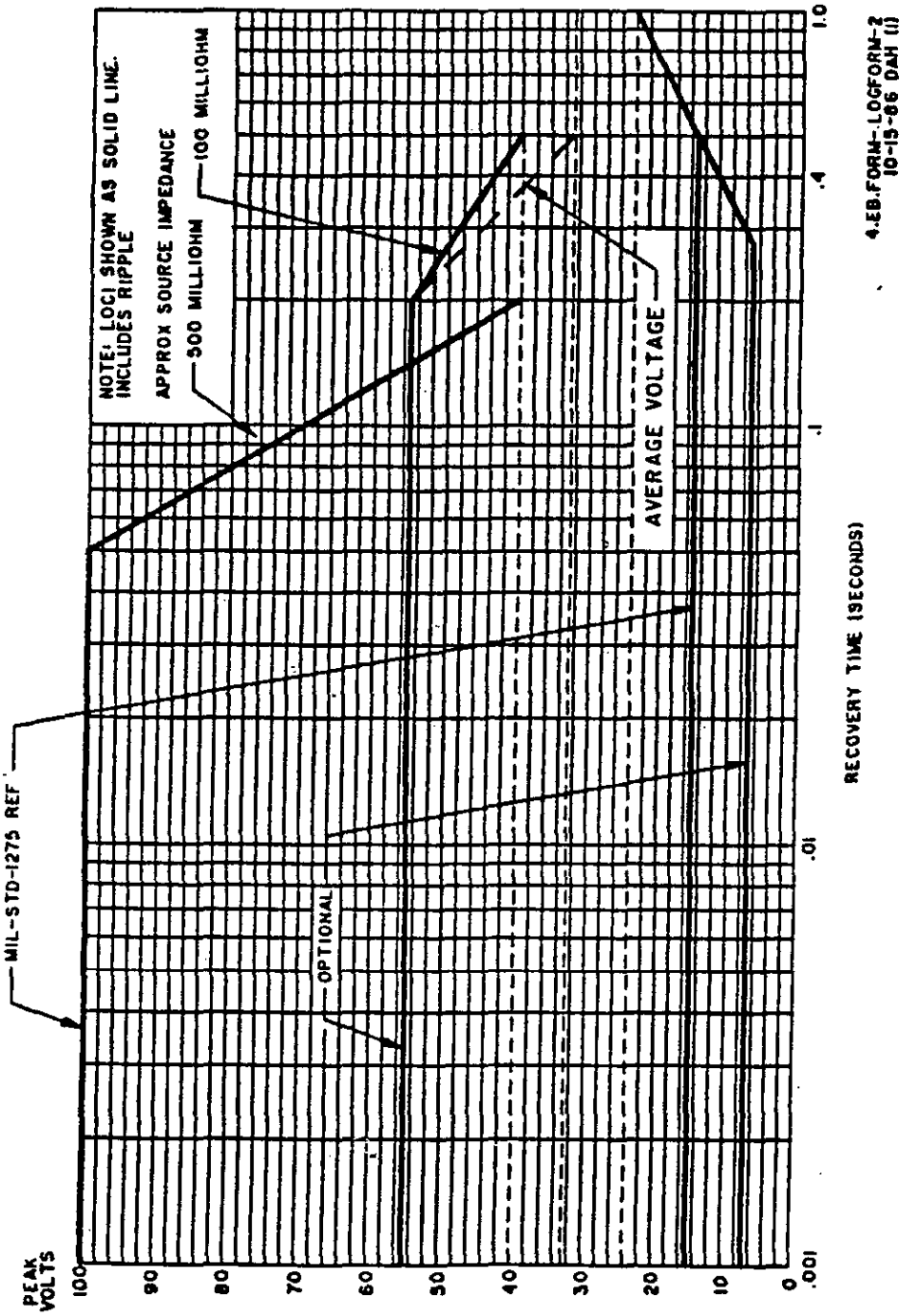


FIGURE 4. Single fault condition, without battery.

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3.5.11 Endurance (see 4.6.1.11). The generator shall be capable of the following 600 hour test without performance degradation of any requirements specified in 3.5.1 through 3.5.10 and without any maintenance or malfunction.

a. Perform a total of 25 cycles as follows:

(1) Three hours loaded to maximum current output at 7000 to 7500 rpm with 100 ± 10°F generator air inlet temperature.

(2) Two hours loaded to 65 ± 5 percent of maximum current output at 7000 to 7500 rpm with 230 ± 10°F generator air inlet temperature.

(3) Seven hours loaded to 65 ± 5 percent of maximum current output at 7000 to 7500 rpm with 180 ± 10°F generator air inlet temperature.

(4) Nine hours loaded to 90 ± 5 percent of maximum current output at 2300 to 3000 rpm with 160 ± 10°F generator air inlet temperature.

(5) Three hours off. Cool down to room temperature.

b. Four times during the 600 hour test (at 120 hours, 240 hours, 360 hours and 480 ± 50 hours), cold soak the generator for 24 hours at -60°F followed by warmup to the room temperature.

3.5.12 Power winding protection (see 4.6.1.12). The generator power windings shall be protected from burnout due to shorted power rectifiers.

3.5.13 Bi-directional operation (see 4.6.1.13). The generator fan shall be capable of operation in both clockwise and counterclockwise directions without adversely affecting the fan's cooling capacity. Electrical and electronic generator components shall not exceed the maximum allowable temperature listed in the contractor's published data sheets when operating at maximum self-limited output at an air inlet temperature of 230°F with the fan operating in either direction.

3.5.14 Parallel operation (see 4.6.1.14). The generator shall be capable of parallel operation with generators conforming to this specification, specification Drawing 19207-12328817, and Drawing 19207-12351001 without additional control or protective devices.

3.5.14.1 Voltage stability and load sharing (see 4.6.1.14). To prevent load share oscillations (hunting) and to ensure load sharing, the generator output voltage shall be stable or decrease with load.

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3.6 Environmental conditions.

3.6.1 Low temperature (see 4.6.2.1). The generator shall demonstrate no performance degradation and show no evidence of damage when stored and operated at -60°F.

3.6.2 High temperature (see 4.6.2.2). The generator shall demonstrate no performance degradation and show no evidence of damage when stored and operated at 230°F.

3.6.3 Shock (see 4.6.2.3). The generator shall demonstrate no performance degradation beyond the limits specified in 4.6.1.1 and shall show no evidence of damage after being subjected to half sine shock pulses of 20g peak value and 62 ± 12ms duration in the vertical direction and 10g and 62 ± 12ms in the longitudinal and transverse direction in accordance with figure 5. Three shock pulses per half axis shall be administered, for a total of 18 shocks.

3.6.4 Vibration (see 4.6.2.4). The generator shall demonstrate no performance degradation beyond the limits specified in 4.6.1.1 and 4.6.1.2 and shall show no evidence of damage during or after being subjected to random vibration in accordance with figure 6.

3.6.5 Humidity (see 4.6.2.5). The generator shall demonstrate no performance degradation beyond the limits specified in 4.6.1.3 and shall show no evidence of corrosion damage after being subjected to a relative humidity ranging from 0 to 98 percent, in accordance with figure 7.

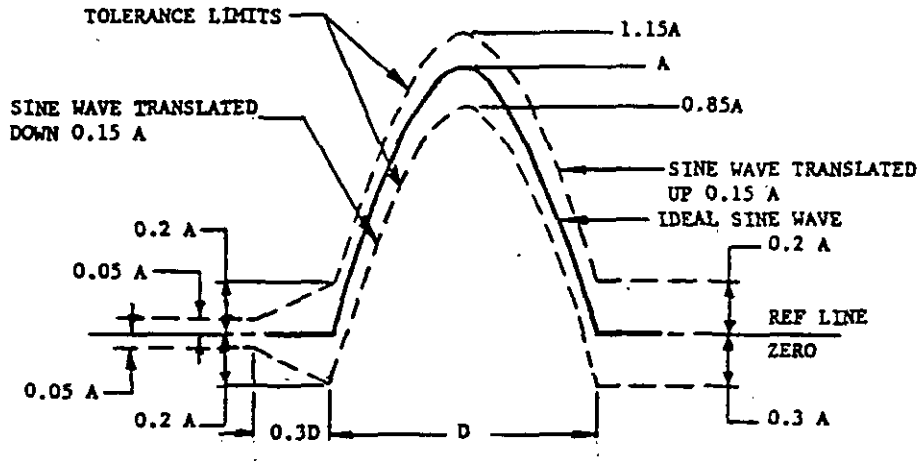
3.6.6 Salt fog (see 4.6.2.6). The generator shall demonstrate no performance degradation beyond the limits specified in 4.6.1.1 after being subjected to salt fog in accordance with MIL-STD-810, method 509.

3.6.7 Steam and waterjet cleaning (see 4.6.2.7). The generator shall demonstrate no performance degradation beyond the limits specified in 4.6.1.1 and shall show no evidence of damage or deterioration following a steam and waterjet cleaning process which uses a cleaner conforming to P-D-220. There shall be no entry of water or moisture to modules sealed against water or moisture.

3.6.8 Fungus (see 4.6.2.8). The generator shall demonstrate no performance degradation beyond the limits specified in 4.6.1.1 and 4.6.1.2 and shall show no evidence of decomposition or deformation after being subjected to a fungus-conducive environment in accordance with MIL-STD-810, method 508.

3.6.9 Dust (see 4.6.2.9). The generator shall demonstrate no catastrophic failure after 288 hours of operation in an extreme dust environment of silica flour with a suspended dust concentration of 5 ounces per cubic foot (oz/ft^3). The silica flour composition shall be as specified in paragraph 4.6.2.9. The dust test shall consist of 12 24-hour cycles. For 20 hours, the generator shall operate at a speed of 6000 ± 200 rpm with an output of 50 ± 5 amps and an air inlet temperature of 200 ± 10°F. During the remaining 4 hours, the generator shall operate without electrical load and shall cool down to 100 ± 25°F.

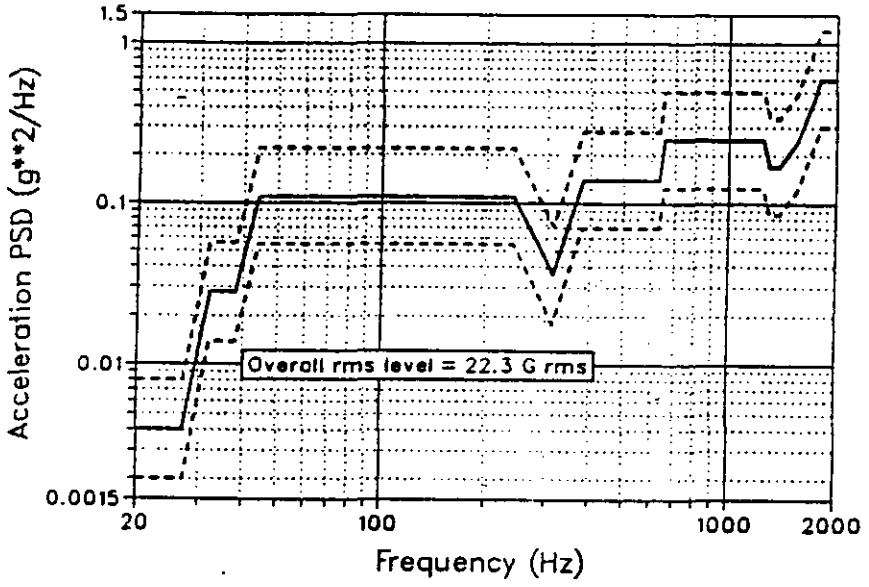
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NOTE: The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The acceleration amplitude of the ideal half sine pulse is A and its duration is D. The measured acceleration pulse shall be contained between the broken line boundaries and the measured velocity change (which may be obtained by integration of the acceleration pulse) shall be within the limits $V_i \pm 0.1\ V_i$ where V_i is the velocity-change associated with the ideal pulse which equals 2 AD/pi. The integration to determine velocity change shall extend from 0.4D before the pulse to 0.1D after the pulse.

FIGURE 5. Half sine pulse configuration and its tolerance limits.

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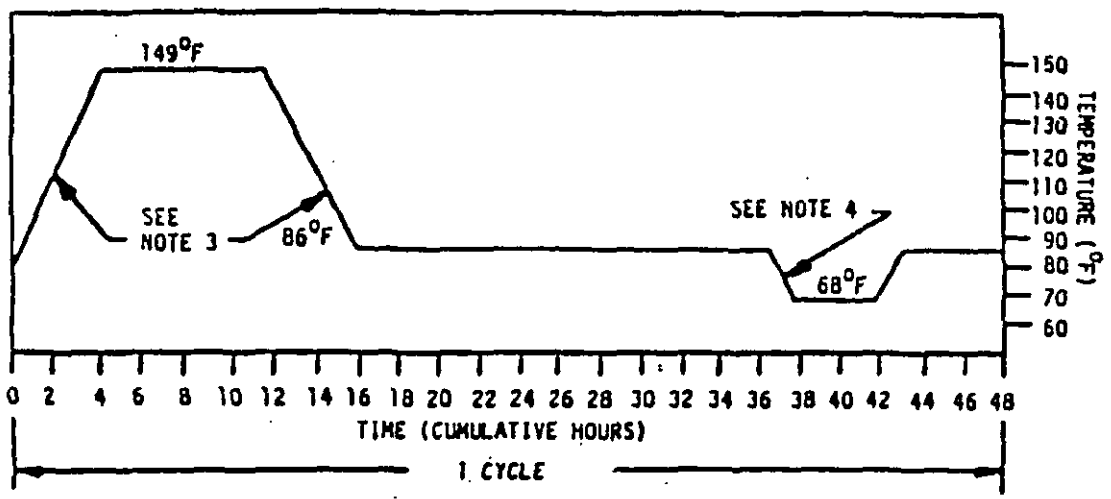
INPUT DATA

Frequency	Acceleration	Frequency	Acceleration
20	.004	630	.140
27	.004	650	.250
32	.028	1250	.250
38	.028	1300	.170
44	.110	1380	.170
240	.110	1560	.250
315	.036	1800	.600
380	.140	2000	.600

NOTE: Generator vibration test specification for the frequency range above 2000 hertz (Hz) consists of a swept sine vibration from 2800 to 2900 Hz at a constant amplitude of 12 g's rms.

FIGURE 6. Random vibration spectrum.

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NOTES:
1. Tolerance during temperature change shall be not greater than 5°F.
2. Relative humidity shall be maintained at 94 ± 4% at all times, except that during the descending temperature period, the relative humidity may be permitted to drop as low as 85%.
3. Rate of temperature change between 86 and 149°F shall be not less than 14.4°F per hour.
4. The temperature increase in this portion of the curve shall be not less than 18°F.

FIGURE 7. Humidity cycle.

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3.7 Product marking. Part identification shall be applied as specified on drawing 19207-12351001.

3.8 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9.

3.9 Interchangeability. To provide for interchangeability, the generator shall conform to this specification and Drawing 19207-12351001. All subassemblies/components selected in design shall make maximum use of military standard and commercially available materials. Use of non-standard materials shall be only in those applications where there is no standard military or commercial material available. Non-standard materials shall be approved by the Government prior to use.

3.10 Electromagnetic radiation (see 4.6.3.1). Conducted and radiated emissions from the generator shall not exceed the CE07 and RE05 levels specified in MIL-STD-461.

3.11 Nuclear survivability (see 4.6.3.2). This is a nuclear hardness critical item. Generator and all solid state (semiconductor) devices used in this assembly must meet the Nuclear Survivability Requirements specified in MRC-82304 MRC/SD-EP-82-027.

3.12 Vehicle system testing (see 4.6.3.3). Vehicle system tests shall be performed to assure generator compatibility with mating system, performance under vehicle operational environment and adequate reliability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to determine conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Materials and manufacturing processes. The contractor shall maintain, and make available for Government review upon request, documentation verifying that materials and processes used in manufacture of the generator conform to specified requirements. As a minimum, documentation shall confirm the following:

a. Materials as specified in 3.2.

b. Soldering as specified in 3.3.6.

c. Power transistor as specified in 3.3.11.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be conducted under the following standard (room) ambient conditions:

a. Temperature: $77 \pm 18^{o}F$

b. Relative humidity: uncontrolled room ambient

c. Atmospheric pressure: site pressure

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4.4 First article inspection.

4.4.1 Preproduction inspection (see 3.1.1). Ten generators produced in accordance with 3.1.1 shall be subjected to preproduction inspection. Two of the generators (samples five and six) shall be built with provisions to test the power winding protection (see 4.6.1.12). These two unique units shall provide the following:

Diode phase lead connections shall be brought out so that they can be connected to B+ or B-, duplicating diode shorts in the positive or negative side of the rectifier bridge.

4.4.1.1 Preproduction inspection routine. The preproduction samples shall be subjected to the inspections specified in table II.

4.4.1.2 Preproduction failure. Inability of any generator to pass any of the tests specified in table II shall be deemed a failure and cause for refusal to grant first article approval. In the event of a failure, the failed part or parts shall be returned to the contractor. The contractor shall perform a failure analysis of each part and submit a corrective action plan of sufficient validity to justify design acceptability without retest. The Government shall approve the corrective action plan and any retest activity.

4.4.1.3 Disposition of preproduction samples. Samples which have been subjected to preproduction inspection shall be indelibly marked "Test Equipment" and shall not be delivered as new equipment under the contract.

4.4.2 Initial production inspection (see 3.1.2). Five generators produced in accordance with 3.1.2 shall be subjected to initial production inspection.

4.4.2.1 Initial production inspection routine. The initial production samples shall be subjected to the inspections specified in table II.

4.4.2.2 Initial production failure. Inability of any of the initial production samples to pass any examination or test shall constitute a failure. Failure of a sample to pass any examination or test shall be cause for refusal to grant first article approval. Such failure may, at the option of the Government, be cause for refusal to allow additional examinations or tests until causes of faults have been corrected and the corrective action approved by the Government.

4.4.2.3 Disposition of initial production samples. Samples which have been subjected to initial production inspection shall be indelibly marked "Test Equipment" and shall not be delivered as new equipment under the contract.

4.4.3 Retention of first article approval. Unless extended by the Government to other contracts, first article approval is valid only on the contract under which it is granted.

4.4.4 Changes to design, processes, procedures, or location. Whenever a change is made in the design or the manufacturing process, procedures, or facility location used in the production of generators, the Government shall be notified to determine whether the contractor must requalify for first article approval.

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TABLE II. First article inspection.

Inspection	Req. para.	Insp. method	Preproduction										Initial Production				
			1	2	3	4	5	6	7	8	9	10	1	2	3	4	5
Group B	table IV	table IV	X	X	X	X	X	X	X	X	X		X	X	X	X	
Weight	3.3.1	Scale	X	X									X	X			
Spline alignment	3.3.4	Visual	X	X									X	X			
Stress screen	3.3.5	4.6.2.10	X	X	X	X	X	X	X	X	X		X	X	X	X	
Sealing of elec. assemblies	3.3.7	4.6.2.11										X					X
Guard	3.3.9	Visual	X	X									X	X			
Diode design reliability	3.3.10	4.6.3.4										X					
Press fit assemblies	3.3.12	4.6.3.5										X					
Generator output current	3.5.1	4.6.1.1	X	X	X	X	X	X	X	X	X		X	X	X	X	
Generator output voltage settings	3.5.2	4.6.1.2.1	X	X	X	X	X	X	X	X	X		X	X	X	X	
Volt. regulation	3.5.3	4.6.1.3.1	X	X	X	X	X	X	X	X	X		X	X	X	X	
Voltage ripple	3.5.4	4.6.1.4	X	X	X	X	X	X	X	X	X		X	X	X	X	
Voltage transients	3.5.5	4.6.1.5	X	X	X	X	X	X	X	X	X		X	X	X	X	
Overvoltage protection	3.5.6	4.6.1.6 through 4.6.1.6.3								X	X				X	X	
Isolated output	3.5.7	4.6.1.7	X	X	X	X	X	X	X	X	X		X	X	X	X	
Efficiency	3.5.8	4.6.1.8	X	X											X	X	
Overspeed	3.5.9	4.6.1.9	X	X	X	X	X	X	X	X	X		X	X	X	X	
Maximum capacity	3.5.10	4.6.1.10	X	X	X	X	X	X	X	X	X		X	X	X	X	

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TABLE II. First article inspection - continued.

Inspection	Req. para.	Insp. method	Preproduction										Initial Production				
			1	2	3	4	5	6	7	8	9	10	1	2	3	4	5
Endurance	3.5.11	4.6.1.11					X	X							X	X	
Power winding protection	3.5.12	4.6.1.12					X	X									
Bi-direct. fan	3.5.13	4.6.1.13	X	X													
Parallel oper.	3.5.14 3.5.14.1	4.6.1.14	X	X													
Low temp.	3.6.1	4.6.2.1	X	X									X	X			
High temp.	3.6.2	4.6.2.2	X	X									X	X			
Shock	3.6.3	4.6.2.3	X	X									X	X			
Vibration	3.6.4	4.6.2.4	X	X									X	X			
Humidity	3.6.5	4.6.2.5	X	X									X	X			
Salt fog	3.6.6	4.6.2.6	X	X									X	X			
Steam & waterjet cleaning	3.6.7	4.6.2.7	X	X									X	X			
Fungus	3.6.8	4.6.2.8	X	X									X	X			
Dust 1/	3.6.9	4.6.2.9	X	X													
Electromagnetic radiation	3.10	4.6.3.1	X	X									X	X			
Nuclear surv. 2/	3.11	4.6.3.2			X	X											
Veh. syst. test	3.12	4.6.3.3							X	X	X						

1/ Dust test shall be performed after electromagnetic radiation.

2/ Functional tests listed for preproduction units 3 and 4 shall be performed before and after nuclear survivability. If schedule permits, preproduction units 1 and 2 may be used. If units 1 and 2 are used, preproduction units 3 and 4 are not necessary; otherwise, preproduction units 3 and 4 shall be used.

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4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections.

4.5.1.1 Group A inspection. Group A inspection shall consist of the inspections specified in table III.

4.5.1.1.1 Group A sampling. 100 percent inspection shall apply to group A inspection.

4.5.1.1.2 Group A failure. The contractor may rework a rejected generator to correct the defect(s) and resubmit it for inspection. Rework shall be within the confines of drawing and specification requirements.

4.5.1.2 Group B inspection. Group B inspection shall consist of the inspections specified in table IV.

4.5.1.2.1 Group B sampling. Frequency of inspection and/or any special arrangements for establishing the frequency of inspection shall be as specified in the contract or purchase order (see 6.2). Recommended requirements for a sampling plan are shown in 6.4.

4.5.1.2.2 Group B failure. The contractor may rework a rejected generator to correct the defect(s) and resubmit it for inspection. Rework shall be within the confines of drawing and specification requirements.

4.5.2 Periodic inspection. Periodic inspection shall consist of groups C and D inspection. Delivery of units which have passed groups A and B inspection shall not be delayed pending completion of periodic inspection.

4.5.2.1 Groups C and D inspection. Groups C and D inspection shall consist of the inspection specified in tables V and VI. Groups C and D inspection shall be made on sample units selected from inspection lots (see 6.6.2) which have passed groups A and B inspection.

4.5.2.2 Group C sampling plan. Unless otherwise specified (see 6.2), sample units shall be selected at the rate of 1 for each 200 units produced.

4.5.2.3 Group D sampling plan. Unless otherwise specified (see 6.2), sample units shall be selected at the rate of 3 for each 600 units produced.

4.5.2.4 Noncompliance. If a sample fails to pass group C or D inspection, the contractor shall notify the Government and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which are manufactured under essentially the same materials and processes, and which are considered subject to the same failure. After the corrective action has been taken, group C and/or D inspection shall be repeated on additional sample units (or the reworked failed unit(s) if so identified and agreed upon in the corrective action plan submitted by the contractor). All tests and examinations, or the test(s)/examination(s) which the original sample failed, shall be performed at the option of the Government. Acceptance and shipment of product shall be discontinued until corrective action, acceptable to the Government, has been taken and group C and D inspections have shown that the corrective action was successful.

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TABLE III. Group A inspection.

Inspection	Requirement paragraph	Inspection method
Generator output current	3.5.1 (tests #2 and #3)	4.6.1.1
Generator output voltage settings	3.5.2	4.6.1.2.2
Voltage regulation	3.5.3	4.6.1.3.2
Overvoltage protection (Group A test only)	3.5.6	4.6.1.6 4.6.1.6.1.2
Isolated output terminal	3.5.7	4.6.1.7
Stress screen	3.3.5	4.6.2.10

TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Inspection method	Classification of characteristics	
			Major	Minor
Spline minor diameter	3.3.2	Go plug gage and not go plug gage	X	
Involute spline dimensions	3.3.2	Involute go composite plug gage 19207-10929031 & Involute not go sector plug gage 19207-10929032	X	
Dimensions	3.3.2	Standard inspection equipment	X	
Protective coating (coverage)	3.4	Visual		X
Product marking	3.7	Visual		X
Workmanship	3.8	Visual	X	

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4.5.2.5 Disposition of sample units. Sample units which have been subjected to group C inspection may be delivered on the contract. Sample units which have been subjected to group D inspection shall be indelibly marked "Test Equipment" and shall not be delivered as new items under the production contract.

TABLE V. Group C inspection.

Inspection	Requirement paragraph	Inspection method
Voltage regulation	3.5.3	4.6.1.3
Voltage ripple	3.5.4	4.6.1.4
Voltage transients	3.5.5	4.6.1.5
Overvoltage protection (Group C test only)	3.5.6	4.6.1.6 4.6.1.6.1.3
Efficiency	3.5.8	4.6.1.8
Overspeed	3.5.9	4.6.1.9
Maximum capacity	3.5.10	4.6.1.10

TABLE VI. Group D inspection.

Sample			Inspection	Requirement paragraph	Inspection method
1	2	3			
X			Endurance	3.5.11	4.6.1.11
	X		Vibration	3.6.4	4.6.2.4
	X		Humidity	3.6.5	4.6.2.5
		X	Sealing of electronic assemblies	3.3.7	4.6.2.11

4.5.3 Inspection of packaging. The sampling and inspection of packaging shall be in accordance with 5.1.

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4.6 Methods of inspection. The following methods of inspection are acceptable for verifying compliance with the section 3 requirements.

4.6.1 Performance tests.

4.6.1.1 Generator output current (see 3.5.1).

a. Connect generator to the test circuit as shown in figure 8.

b. Close switches S1, S3 and S4 to connect battery and load.

c. Operate generator for a period of not less than 1 hour at each of the specified speed and temperature conditions of table I. Measure and record output current and verify that it meets values specified in table I. Verify that the excitation current does not exceed 1 amp. For 100 percent inspection (Group A), only tests #2 and #3 shall be performed.

4.6.1.2 Generator output voltage settings (see 3.5.2). To verify performance requirements of paragraph 3.5.2, perform the following test as applicable.

4.6.1.2.1 First article, group C and group D tests.

a. Remove the regulator from the generator and place in a temperature chamber.

b. Connect the generator to the test circuit as shown in figure 8. With S1, S3 and S4 closed, operate the generator at 7500 $\pm$ 250 rpm with battery and a 50 $\pm$ 5 amp load.

c. Stabilize the regulator temperature at -60, 0, 75, 150, and 250°F.

d. After regulator stabilization at each test point, verify the generator voltage setpoints are in accordance with figure 2.

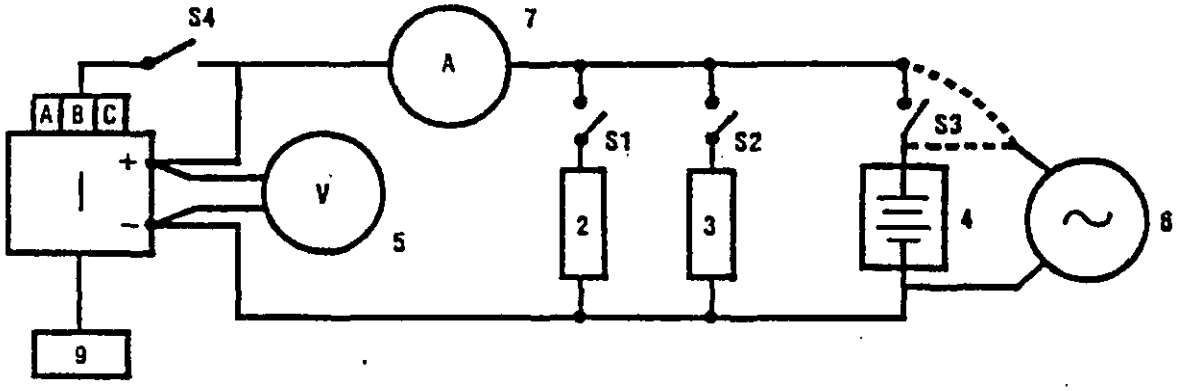
4.6.1.2.2 Group A test. Perform the following test at room temperature.

a. With the regulator mounted on the generator, attach a thermocouple to the regulator housing and connect the generator to the test circuit shown in figure 8.

b. With S1, S3, and S4 closed, operate the generator at 7500 $\pm$ 250 rpm with battery and a 50 $\pm$ 5 amp load.

c. After temperature stabilization, verify the generator output voltage is in accordance with figure 2 for the measured regulator temperature.

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LEGEND:

1. Generator under test
2. Adjustable resistive load bank
3. Auxiliary load bank
4. Battery 200AH fully charged
5. Voltmeter (4 digit minimum)
6. Oscilloscope
7. Ammeter (400 A)
8. S1, S2, S3, S4 switches
9. Tach (motor/rpm monitor)

NOTE: Switch S3 must be closed for field excitation to start generator output. After output is started, S3 may be opened and output will continue.

FIGURE 8. Generator Test Set-Up.

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4.6.1.3 Voltage regulation (see 3.5.3). Perform the following test as applicable.

4.6.1.3.1 First article, group C and group D tests.

a. With the generator test setup and operation as specified in 4.6.1.2.1, adjust the load to 10, 50 and 90 percent of maximum output current. Verify the generator voltage regulation is in accordance with figure 2.

b. Change generator speed to 2350 ± 50 rpm. Perform the test as above, but load the generator to 25, 50 and 90 percent of maximum output current. Verify the voltage regulation is in accordance with figure 2.

c. Disconnect the battery.

d. Repeat above tests and verify the voltage tolerance is within ± 1.0 vdc from readings obtained in 4.6.1.2.1. Verify internal sensing by inserting a silicon diode (0.65 to 0.70 vdc drop) in the energizing lead. The output voltage of the generator shall not change. Perform this test once at 7500 rpm and 10 percent load.

4.6.1.3.2 Group A test. Perform the following test at room temperature.

a. With the generator operating as specified in 4.6.1.2.2, adjust the load to 10, 50 and 90 percent of maximum current output.

b. Verify the output voltage is in accordance with figure 2 for the measured regulator temperature.

4.6.1.4 Voltage ripple (see 3.5.4).

a. Connect the generator to the test circuit shown in figure 8.

b. Close switches S1, S3 and S4 to energize the generator and connect battery and load.

c. Operate the generator at 2350 rpm at 25, 50 and 90 percent of maximum current output.

d. Operate the generator at 7500 rpm at 10, 50 and 90 percent of maximum current output.

e. Verify p-p ripple voltage does not exceed 3.0 vdc when measured with an oscilloscope at the battery terminals.

f. Disconnect the battery (open switch S3), connect the oscilloscope to the output terminals of the generator.

g. Verify peak output voltage including ripple does not exceed 40.0 vdc.

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4.6.1.5 Voltage transients (see 3.5.5).

a. Connect generator to the test circuit shown in figure 8.

b. Close switches S2 and S4 to energize the generator and to connect an auxiliary load of 10 percent of rated output.

c. Close switch S1 and adjust the main load for a total current of 85 percent of rated continuous current rating.

d. Open switch S1 and measure the voltage transient on the oscilloscope. The voltage transients shall be within the loci of figure 4. Switching time (arc duration) as observed on the oscilloscope shall not exceed 20 ms. A spring loaded knife switch shall be used. Capacitors to adjust the switching time are not allowed.

e. Repeat steps a. through d., except close switch S3 to connect the battery. A fully charged battery shall be used (battery shall draw less than 5 amps from a 28 volt charging source). All surges and spikes shall be within the loci of figure 3.

f. Repeat steps a. through d. by switching the load from 10 percent to 85 percent with battery and without battery. All surges and spikes shall be within the loci of figure 3 and figure 4, respectively. The voltage transient tests shall be performed at a generator speed of 7000 to 7500 rpm.

The voltage transient test shall not cause generator shutdown. Repeat steps a. through d. by operating 2 generators in parallel as shown in figure 9. The rated continuous current is the combined generator 1 and generator 2 output current.

4.6.1.6 Overvoltage protection (see 3.5.6). Test generators shall be modified with a switch to defeat the generator voltage control. Without voltage control, the generator output voltage rises until the overvoltage protection shuts the generator down. After each shutdown test, the circuit shall be reset or restored as required. Shutdown time from start of failure simulation (switch operations) to generator shutdown and peak voltage shall be in accordance with the requirements specified in 3.5.6.1 and 3.5.6.2. Subject generator to the following test(s) as applicable.

4.6.1.6.1 Overvoltage shutdown test, single generator operation. Perform the following test as applicable.

4.6.1.6.1.1 First article test. Perform the following test with test generator at the regulator/overvoltage circuit temperatures of -60°F, 75°F and 250°F.

a. Connect test generator to the test circuit as shown in figure 8, generator 1.

b. Operate the generator with and without battery at 2350 rpm and 7500 rpm at 25 percent and 90 percent of the maximum generator output.

c. Initiate the overvoltage condition (see 4.6.1.6).

d. Record the peak voltage and shutdown time.

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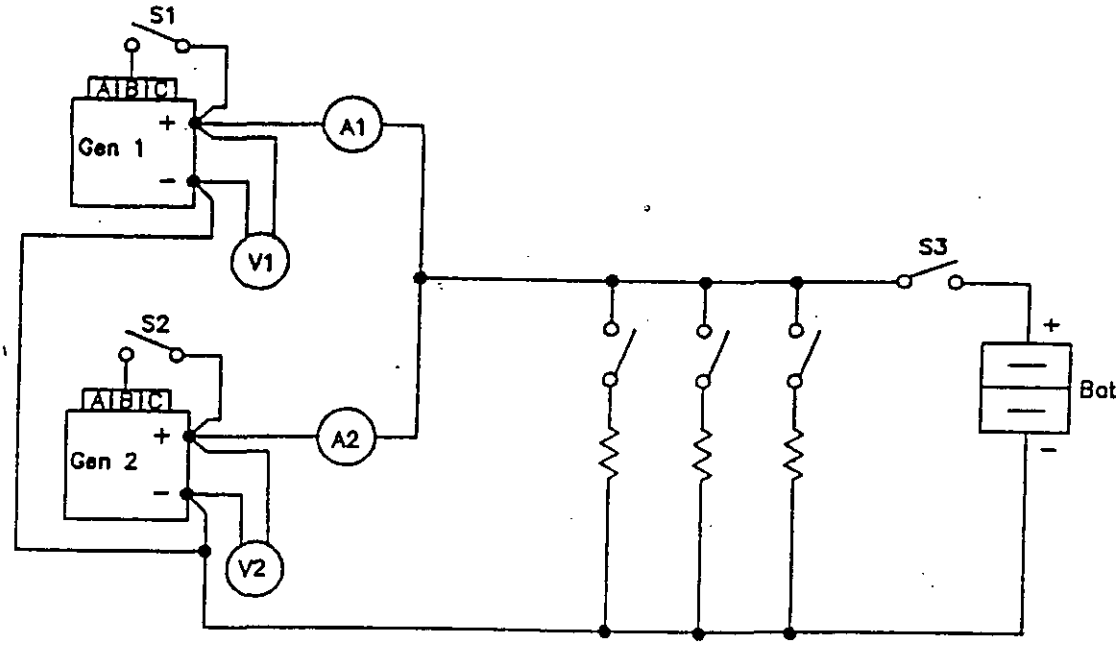


FIGURE 9. Parallel operation test setup.

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4.6.1.6.1.2 Group A test.

a. Connect a resistor across the output terminals of the power transistor controlling the field.

b. Apply a current limited dc power supply to the output terminals of the generator; monitor the power supply current.

c. Increase the voltage to 31.0 ± 0.5 vdc. The overvoltage circuit shall not trip.

d. Increase the voltage to 35.0 ± 0.5 vdc. The overvoltage circuit shall trip.

4.6.1.6.1.3 Group C test.

a. Connect test generator to the circuit as shown in figure 8, generator 1.

b. Operate generator with battery at 7500 rpm and 25 percent load.

c. Initiate the overvoltage condition (see 4.6.1.6).

d. Record the peak voltage and shutdown time.

4.6.1.6.2 Overvoltage shutdown test, parallel generator operation (first article).

a. Connect two test generators to the test circuit as shown in figure 9, generators 1 and 2.

b. Perform the shutdown test with battery at 2350 rpm and 7500 rpm at 25 percent and 90 percent of the combined (generator 1 and generator 2) maximum output current.

c. Perform this test by initiating the overvoltage condition in generator 1. Only generator 1 shall shut down. Generator 2 shall be operational.

d. Repeat the test by initiating the overvoltage in generator 2. Only generator 2 shall shut down. Generator 1 shall be operational.

4.6.1.6.3 Overvoltage protection, immunity to shutdown, 100 vdc spike test (first article).

a. Connect test generator to the test circuit as shown in figure 9, generator 1.

b. Operate the generator at 7500 rpm without battery or external load.

c. Apply 5 100-vdc pulses with 30 ms duration at 1 second intervals to the generator B+ terminal and ground. The source impedance of the 100-vdc pulse shall be not greater than 10 ohms.

d. To verify that the generator is functional, apply a load of 90 percent of the maximum generator output. The generator output voltage shall be per figure 2.

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4.6.1.7 Isolated output terminal (see 3.5.7).

a. Connect generator to the test circuit as shown in figure 8.

b. Operate the generator at 7500 rpm.

c. Close S3, S4 and S1 to energize and connect the battery. Apply a load of 10 percent of maximum generator output. The voltage on terminal C of the energizing receptacle shall be within ± .5 vdc of the B+ output terminal voltage.

d. Turn off S4. The voltage at terminal C shall be less than 10 vdc.

4.6.1.8 Efficiency (see 3.5.8). Remove the temperature-compensated regulator from the generator and hold at a temperature of less than 80°F before performing the following test.

a. Measure the generator input torque and rpm between 65 and 85°F ambient temperature while the generator produces maximum output current at 28.0 vdc. Measure the generator torque and output current in the 2000 to 8500 rpm speed range and record after the machine temperature has stabilized (less than 1 amp output current change in 5 minutes).

The generator input power (P [W_i])

$$P\ [W_i] = \text{Torque}\ \frac{(\text{inch lbs.})}{84.6} * \text{rpm}$$

The generator output power (P [W_o])

$$P\ [W_o] = I(\text{out}) * V(\text{out})$$

$$\text{The efficiency [\%]} = \frac{P\ (\text{output})}{P\ (\text{input})} * 100$$

b. Verify that peak torque does not exceed 40 foot pounds and the efficiency meets the requirements of 3.5.8.

4.6.1.9 Overspeed (see 3.5.9).

a. Operate the generator for a minimum of 60 seconds at a speed of 10,000 rpm with a load of 0 to 15 amp. While monitoring the output voltage, there shall be no indication of any damage.

b. Adjust the load to 90 percent of maximum current output and operate the generator at 7000 to 7500 rpm. Compare the output voltage with figure 2 to verify that there is no impairment of generator performance.

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4.6.1.10 Maximum capacity (see 3.5.10).

a. Connect the generator to a test circuit shown in figure 8.

b. Place the generator in a temperature chamber and control generator air inlet temperature at 230 ± 5°F. Adjust load during the test to maintain 26.0 ± 0.5 vdc output voltage.

c. Operate the generator for 6 hours at 2350 ± 50 rpm and 6 hours at 7250 ± 250 rpm.

d. Perform the voltage output test (see 4.6.1.3.2) before and after maximum capacity test to verify that there is no impairment of generator performance.

4.6.1.11 Endurance (see 3.5.11). Subject generator to 600 hours endurance test as specified in 3.5.11. No maintenance of any kind shall be allowed during the test. At the completion of the endurance test, subject the generator to the current output, voltage output and voltage regulation tests as specified in 4.6.1.1 through 4.6.1.3.2.

4.6.1.12 Power winding protection (see 3.5.12). Perform the following steps.

a. Short-circuit one of the diodes connected to B+ (see 4.4.1).

b. Energize the generator.

c. Operate the generator in accordance with table VII. Terminate tests 1 through 6 once fuse opens. Remove short. Replace fuse.

d. Repeat steps a., b. and c. shorting a diode connected to B-.

e. Disassemble as required to expose windings. Windings shall show no evidence of damage or deformation.

TABLE VII. Power winding protection.

Test	Speed (rpm)	Load (amps)	Time (minutes)
1	2350	15	15 ± 2
2	2350	50	15 ± 2
3	7500	15	15 ± 2
4	7500	50	15 ± 2
5	7500	100	15 ± 2
6	7500	200	15 ± 2

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4.6.1.13 Bi-directional operation (see 3.5.13).

a. Connect a thermocouple to the base of three positive and three negative diode cases per diode manufacturer's recommended practice.

b. Operate the generator at 2350, 4000 and 7500 rpm with an air inlet temperature of 230°F.

c. Apply load until the generator output voltage drops to 26 vdc.

d. Allow the diode temperature to stabilize.

e. Perform this test in the clockwise and counterclockwise directions. The maximum diode temperature for generator operation in each direction shall be within the diode manufacturer's published data sheet limits.

4.6.1.14 Parallel operation (see 3.5.14).

a. Connect test generators 1 and 2 to the test circuit shown in figure 9.

b. Perform the voltage regulation test as specified in 4.6.1.3 on generator 1 by closing S1 and on generator 2 by opening S1 and closing S2. Verify that, with load increase, the output voltage of both generators is either stable or decreases with load increase.

c. Operate the generators in parallel by closing switches S1 and S2. Set the speed of both generators to 7500 rpm. Increase the load to 90 percent of the combined maximum current output of both generators. Verify the current output of either generator is not more than 3 times the current output of the other generator. Perform this test with and without batteries.

d. With a single switch, remove and reconnect a load of 100 amps $\pm$ 20 percent. Verify there are no voltage or current oscillations. Perform this test with and without batteries.

4.6.2 Environmental tests. The following methods of inspection are acceptable for verifying compliance with 3.6. Prior to and at the conclusion of each of the environmental tests, the generator shall be operated under stabilized standard ambient conditions (see 4.3) and examined for evidence of deformation, damage and performance degradation.

4.6.2.1 Low temperature (see 3.6.1). Place the generator in a temperature chamber. Lower chamber temperature to -60°F and maintain for a period of 24 hours. Remove generator from temperature chamber. Within 5 minutes after removal, operate the generator at 2350 rpm at 25, 50 and 90 percent of maximum current output, and at 7500 rpm at 10, 50 and 90 percent of maximum current output. Verify the voltage regulation is in accordance with figure 2.

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4.6.2.2 High temperature (see 3.6.2). Connect the generator to the test circuit as shown in figure 8 and place generator in a temperature chamber. Raise chamber temperature to 230°F and maintain for a period of 48 hours with a relative humidity not greater than 15 percent. During high temperature exposure, operate the generator at 2350 rpm at 25, 50, and 90 percent of maximum current output, and at 7500 rpm at 10, 50 and 90 percent of maximum current output. Verify that voltage regulation is in accordance with figure 2.

4.6.2.3 Shock (see 3.6.3). Mount generator in a test fixture and subject to a shock test as specified in 3.6.3. No operational tests are required during shock pulses. At the conclusion of the shock test, subject the generator to the current output test as specified in 4.6.1.1 for a period of 10 minutes at each of the specified speeds and temperatures.

4.6.2.4 Vibration (see 3.6.4). The generator shall be mounted in a test fixture and subjected to the vibration test specified in 3.6.4. The generator shall be subjected to the vibration spectrum for 3 hours minimum in each of three mutually perpendicular axes, with the first axis of vibration being the shaft. During vibration testing, the generator shall be operated at single speed between 2000 and 7000 rpm with a load of 15 amps. The vibration spectrum shall be maintained within ± 3 dB of the input level, as shown on figure 6. Also during vibration, the output shall be within ± 0.6 vdc of the nominal voltage. At the conclusion of the vibration test, the generator shall be subjected to the current output and voltage output tests as specified in 4.6.1.1 and 4.6.1.2.

4.6.2.5 Humidity (see 3.6.5).

a. The generator shall be placed in a chamber trap-vented to the atmosphere to prevent the build up of total pressure. No condensation shall be permitted to drip on the generator. Relative humidity shall be determined from dry bulb/wet bulb thermometer comparison method or an equivalent method approved by the Government. Air velocity flowing across the wet bulb shall be not less than 900 feet per minute (ft/min). Provisions shall be made for controlling the flow of air throughout the internal chamber test space where the velocity of air shall be not greater than 150 ft/min. Steam or distilled, demineralized, or deionized water having a hydrogen ion concentration (PH) between 6.0 and 7.2 at 73°F shall be used to obtain the specified humidity. No rust or corrosive contaminants shall be imposed on the generator by the test facility.

b. After having been dried for 24 hours at 129 ± 3°F, the generator shall be conditioned at 73 ± 3°F and 50 ± 10 percent relative humidity for 24 hours.

c. Raise the internal chamber temperature to 86 ± 3°F and relative humidity to 94 ± 4 percent. Expose the generator to 5 continuous 48-hour cycles in accordance with figure 7. Accumulated moisture may be removed by turning the generator upside down or shaking. Wiping is not permitted. After completion of cycling, condition the generator for 24 hours at 73 ± 3°F and 50 ± 10 percent relative humidity. Immediately upon removal of generator from humidity chamber, place generator in a moisture proof container, to remain therein until beginning of performance tests. Within 1 hour after humidity testing, subject the generator to the voltage regulation test as specified in 4.6.1.3.

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4.6.2.6 Salt fog (see 3.6.6). Subject the generator to salt fog testing as specified in 3.6.6. No operational tests are required during salt fog testing. After salt fog exposure, subject the generator to the current output test as specified in 4.6.1.1 for a period of 10 minutes at each of the specified speeds and temperatures.

4.6.2.7 Steam and waterjet cleaning (see 3.6.7). Subject the generator to steam cleaning as specified in 3.6.7. Immediately after steam cleaning, subject the generator to waterjet cleaning as specified in 3.6.7. Within .50 hour after completing the steam and waterjet cleaning, subject the generator to the current output test as specified in 4.6.1.1 for a period of 10 minutes at each of the specified speeds and temperatures. Jet pressure shall be 105 ± 5 pounds per square inch gage (psig) for steam and 50 ± 5 psig for water. The jet shall be applied perpendicular to the generator surface from a distance of not more than 1 foot for steam and not more than 3 feet for water, and moved at such a rate that the exposed surface of the generator is subjected to the jet at the rate of 1 square foot per minute. Visually examine modules sealed against water or moisture to verify that no water or moisture has entered.

4.6.2.8 Fungus (see 3.6.8). Subject the generator to a fungus test as specified in 3.6.8. The presence of loosely spread microbial colonies on substrate surfaces is permissible (grade 2). There shall be no evidence of decomposition or deformation. After the fungus test, subject the generator to the current output and voltage output tests as specified in 4.6.1.1 and 4.6.1.2.

4.6.2.8.1 Optional method. If the responsible Government agent grants approval, the fungus testing requirement of 3.6.8 may be met by engineering analysis, previous testing or certification that materials used are of a nonnutrient base.

4.6.2.9 Dust (see 3.6.9).

a. Mount the generator in a dust chamber with the shaft axis horizontal and subject to the dust test specified in 3.6.9, using silica flour per MIL-STD-810 containing 97 to 99 percent (by weight) silicon dioxide (SiO_2). The silica flour dust shall be capable of passing through mesh screens in percentages as specified in table VIII. Use a fan to circulate the air vertically in the chamber and keep the dust suspended. The humidity in the chamber shall be not greater than 30 percent and the air temperature shall be 200 ± 10°F.

TABLE VIII. Dust size distribution.

Percent	Screen mesh size
100	100
98 ± 2	140
90 ± 2	200
75 ± 2	325

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b. Start cycle 1 by operating the generator at 6000 ± 200 rpm at a load of 50 ± 5 amps. Turn on the chamber heater to bring the generator air inlet temperature to 200 ± 10°F.

c. After 20 hours of generator operation, turn off the generator load and chamber heater to allow the air inlet temperature to drop to 100 ± 25°F. Operate the generator without heat and load for 4 hours.

d. Repeat step b. and c. four times.

e. Remove and steam clean the generator and operate the generator for 1 hour at 6000 rpm with a load of 50 ± 5 amps.

f. Replace the dust in the dust chamber and reinstall the generator in the dust chamber.

g. Repeat steps b. through f.

h. Repeat steps b. and c. two times.

i. Remove and steam clean the generator.

j. Subject the generator to current output test #2 of table I as specified in 4.6.1.1, except maintain a minimum output current of 150 amp. If the generator rotor strikes the stator or a torque of more than 15 inch-pounds is required to turn the rotor (generator not energized), the generator is considered to have failed.

4.6.2.10 Stress screen (see 3.3.5).

4.6.2.10.1 Stress screen for the control unit(s). Install control unit(s) into the temperature chamber. The following steps shall be performed:

a. Apply 31 ± 1 vdc to the control unit(s) terminal (B+) and ground. Install a 100 ohm ± 10 percent, 10 watt, resistor from B+ to the collector of the field power transistor.

b. Monitor the current drain at the first and last stress screening cycles at maximum temperature.

c. Perform 15 stress screen cycles as shown in figure 1. Maximum temperature shall be 250 +0 -7.5°F, minimum temperature shall be -65°F (-0 +5°F).

d. Remove power and remove control unit(s) from the chamber.

e. Compare the current drain of the first cycle to the last cycle. Verify that the current drain of the last cycle is within 25 percent of the current drain of the first. If the verification points successfully pass all above steps, indelibly mark the control unit(s) with a yellow dot.

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4.6.2.10.2 Stress screen for the rectifier assembly. Install the rectifier assembly into the temperature chamber. The following steps shall be performed:

a. Apply 31 ± 1 vdc to the rectifier (connect to B+ cable connect and ground (GND) cable connect).

b. Monitor the current drain at the first and last stress screening cycles.

c. Perform 10 stress screen cycles as shown in figure 1. Maximum temperature shall be 300°F (+0 -10°F), minimum temperature shall be -60°F.

d. Remove power and remove rectifier from the chamber.

e. Compare the current drain of the first cycle to the last cycle. Verify that the current drain of the last cycle is within 25 percent of the current drain of the first. If the verification points successfully pass all steps, indelibly mark the control unit(s) with a yellow dot.

NOTE: For a one-time determination of actual time required, connect a thermocouple to control unit(s)/rectifier sample. The chamber shall be held constant at both high and low temperature extremes until the assembly temperature is within 5°F of the chamber temperature. Add 5 minutes to the time that it takes to stabilize within 5°F. This time shall become T1 and T2 (see figure 1).

4.6.2.11 Sealing of electronic assemblies (see 3.3.7).

a. Remove electronic assembly from generator. If the electronic assembly is inseparable from the generator, a special assembly may be built for the sealing test which duplicates the electronic assembly.

b. Cycle test assembly five times between -60°F and 250°F. The rate of temperature change shall be not greater than 10°F per minute.

c. Connect the test assembly to the generator with an extension cable. Operate the generator at 2350 rpm at a 50 amp load while performing steps d. through g.

d. Submerge assembly in a boiling, 5 ± 1 percent salt water solution for not less than 10 minutes.

e. Remove assembly from boiling water and immediately submerge in 80°F maximum, 5 ± 1 percent salt water solution for not less than 10 minutes.

f. Repeat steps d. and e. two more times.

g. Rinse test assembly in fresh water.

h. Perform the voltage transients test per 4.6.1.5 and verify the assembly meets the requirements specified therein.

i. Disconnect test assembly and reinstall original assembly.

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4.6.3 Special tests

4.6.3.1 Electromagnetic radiation (see 3.10). The generator shall be tested in accordance with MIL-STD-462. The test is performed at standard ambient temperature. The machine temperature need not be stabilized. Make conducted and radiated measurements at 2300 to 2400 rpm at 50 and 100 percent of rated output current; make measurements at 7000 to 7500 rpm at 50 and 100 percent of rated output current, and at maximum output current. To obtain maximum output current, the generator shall be loaded until the generator terminal voltage drops to 27.0 ± 0.3 vdc. Verify that the conducted and radiated emission does not exceed the CE07 and RE05 levels in accordance with MIL-STD-461A.

4.6.3.2 Nuclear survivability (see 3.11). Conformance to the requirements of 3.11 shall be verified through testing or engineering analysis as directed by the responsible Government agent.

4.6.3.3 Vehicle system testing (see 3.12). Use at least three of the preproduction generators for vehicle systems testing, including one that will be a spare. Install at least two, and no more than four, generators on vehicles and operate for a combined total of at least 12,000 miles in conditions specified for reliability in the product function specifications for the vehicles. Baseline test each generator in accordance with tables III, IV and V before vehicle tests to assure conformance to performance and drawing requirements. After completion of the required vehicle mileage, subject generator to the current output, voltage output and voltage regulation tests as specified in 4.6.1.1, 4.6.1.2.2 and 4.6.1.3.2. Vehicle testing shall be the responsibility of the Government, whereas tables III, IV and V testing shall be performed by the contractor.

4.6.3.3.1 Reliability requirements. Minimum acceptable reliability as demonstrated during vehicle testing shall be 6000 mean miles between failure (MMBF). Reliability calculation shall be a point estimate based on total vehicle test miles divided by the total chargeable failures attributed to the generator system. Failure shall be defined herein as generator or regulator replacement required because of inability to perform its required function.

4.6.3.4 Diode design reliability. Power rectifier requirements specified in 3.3.10 shall be verified during qualification testing. The following measurements are required:

a. Measure and record the actual forward voltage drop of each power diode used in the generator to be tested (use test method and parameter per diode manufacturer V_{FM} measurement).

b. Measure and record the case temperature of each diode at stabilized maximum generator output at 2350, 5000 and 7500 rpm (use the diode manufacturer recommendation for thermocouple location and attachment method).

c. Measure and record output current, output voltage, ambient and generator air inlet temperature at each test point in b. above.

The above information and the diode manufacturer's specification sheet shall be included in the test report.

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4.6.3.5 Press fit assemblies (see 3.3.12). Perform this test for each unique press fit assembly contained in the generator. One set of components per each unique press fit assembly shall be selected (or fabricated, if necessary) to ensure maximum interference, and 1 set shall ensure minimum interference. Each component set shall be soaked at each service temperature extreme and then immediately placed in a test fixture. A force shall be gradually applied until the force as specified in 3.3.12 is obtained; any relative movement of the components shall constitute a failure.

5. PACKAGING

5.1 Preservation, packaging and marking. Packaging for the desired level of protection shall be in accordance with MIL-P-116 or packaging data sheet specified by the Government (see 6.2).

6. NOTES

6.1 Intended use. Generators manufactured under this specification are intended for use on military tracked vehicles in the Bradley Fighting Vehicle and Fighting Vehicle System Carrier families.

6.2 Ordering data. Procurement documents must specify the following:

a. Title, number, revision, and date of this specification.

b. Title, number, revision, and date of part drawing.

c. Issue of the DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

d. Preproduction inspection, if required (see 3.1.1).

e. Initial production inspection, if required (see 3.1.2).

f. Responsibility for inspection, if other than as specified (see 4.1).

g. Sampling plan requirements for group B inspection (see 4.5.1.2.1).

h. Periodic sampling plan, if other than specified (see 4.5.2.2 and 4.5.2.3).

i. Test sample disposition, if other than specified (see 4.5.2.5).

j. Level of packaging (see 5.1).

k. Any deviation from this specification.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to bidders whether it should include preproduction samples and/or initial production samples (see 3.1), and the number of generators to be tested as specified in 4.4. The acquisition document should

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include specific instructions regarding the arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive first article inspection, or any portion thereof, to those bidders offering a product which has been previously acquired or tested by the Government. Bidders offering such products should furnish evidence with the bid that prior first article approval is presently appropriate for the pending contract.

6.4 Recommended group B sampling plan (see 4.5.1.2.1).

6.4.1 Basic method. Quality characteristics listed in table IV should be inspected using the sampling plan shown in table IX. Alternate sampling plans are not allowed except under an approved statistical process control (SPC) system conforming to the requirements outlined in paragraph 6.4.2.

6.4.1.1 Tightened inspection. Tightened inspection should be introduced as soon as 2 out of 5 successive lots have been rejected and should as a minimum, impose a 30 percent increase in sample size. Normal inspection sampling may be restored after 5 successive lots have been accepted under tightened inspection.

6.4.1.2 Reduced inspection. Reduced inspection may be introduced when 10 successive lots have been accepted and should as a maximum, permit a 30 percent decrease in sample size. Normal inspection sampling should be restored if a lot is not accepted under reduced inspection.

TABLE IX. Group B sampling plan.

Lot Size	Sample Size	
	For Major Characteristics	For Minor Characteristics
2 to 13	Entire lot	3
14 to 25	13	3
26 to 50	13	5
51 to 90	13	6
91 to 150	13	7
151 to 280	20	10
281 to 500	29	11
501 to 1200	34	15
1201 to 3200	42	18
3201 to 10,000	50	22

NOTE: Accept lot on zero defects. Reject lot on one or more defects.

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6.4.2 Statistical process control (SPC) method. Sampling plan requirements of 6.4.1 may be satisfied through the use of SPC as an alternate control method. An SPC plan, detailing the methods for monitoring process control, should be approved by the Government or agent. The plan, developed by the contractor, should include as a minimum:

a. Types of control charts and their uses.

b. Process capability (C_{pk}) studies.

c. Criteria for determining out-of-control conditions.

d. Corrective actions to be taken if an out-of-control or out-of-tolerance condition is detected.

e. Training programs and qualification requirements of personnel executing the plan.

f. Provisions for reduced inspection.

The plan should contain the results of process capability studies for the processes being controlled. For variable data, the minimum C_{pk} for statistical control should be 1.33. For attribute data, a minimum process average of 99.73 percent is required.

6.5 Voltage transient decay time (see 3.5.5). MIL-STD-1275 shows a typical generator impedance of 500 milliohms, indicating that the tests performed to establish the transient voltage decay time were run on a 60 amp generator. A 300 amp generator with a typical impedance of 100 milliohms and increased field inductance requires a longer transient voltage decay time. The optional surge voltage profile as shown in figure 4 applies only 55 vdc to connected loads and although the total time duration exceeds MIL-STD-1275 specification, the surge energy (proportional to V^2 * time) supplied to connected loads does not exceed the surge energy per MIL-STD-1275.

6.6 Definitions.

6.6.1 Generator. The engine accessory generator is the prime source of 28 vdc electrical energy for military vehicles. Generator output is in accordance with table I.

6.6.2 Inspection lot. An inspection lot consists of all units produced under essentially the same conditions and offered for inspection at one time.

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6.7 Subject term (key word) listing.

Overvoltage
Potted assemblies
Power rectifier
Power transistor
Power winding
Spline
Temperature compensation
Voltage: regulation, ripple, transients

Custodian:
Army - AT

Preparing activity:
Army - AT

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-G-62743(AT)	92/09/17

3. DOCUMENT TITLE
Generator, Engine Accessory, 300 AMP

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial: (313) 574-5413 (2) AUTOVON: 786-5413
c. ADDRESS (Include Zip Code) Commander U.S. Army Tank-Automotive Command ATTN: AMSTA-GDS Warren, Michigan 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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